

NATIONAL INSTITUTE OF TECHNOLOGY ROURKELA
DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING
SUBJECT: Foundations of Computer Security
End-Semester Examination for B.Tech Only

■ CS 6301 ■ Duration 180 minutes ■ Full Marks 50 ■ Date: 22-November-2024■

Answer ALL Questions

All parts of the question must be answered at one place.

Number within square bracket indicate full mark for that question.

1. (a) In RSA given n and $\Phi(n)$, calculate the value of p and q . [2]
(b) Given the super increasing tuple $b = [7, 11, 23, 43, 87, 173, 357]$, $r = 41$ and modulus $n = 1001$, encrypt and decrypt the letter 'c' using knapsack crypto-system. Use $[7, 6, 5, 1, 2, 3, 4]$ as permutation table. [3]
(c) Alice uses Bob's RSA public key ($e=7, n=143$) to send the plain-text $P=8$ encrypted as ciphertext $C=57$. Show how Eve can use chosen ciphertext attack if she has access to Bob's computer to find the plain text. [5]
2. (a) Given $g=11$, $p=61$, $x=7$, and $y=3$ for the Diffie-Hellman. Show all the steps in getting the shared secret key? [2]
(b) Decrypt the message "ABXQAYAQTANAFOY" using affine cipher with keys (3, 14)? [3]
(c) In ElGamal given the prime $p = 31, e_1 = 3$ and $d = 10$. [5]
 - i. Find e_2 .
 - ii. Encrypt the message "CRYPTO" use 00 to 25 for encoding. Use Different blocks to make $P < p$.
 - iii. Decrypt the cipher text to obtain the plain text
3. (a) Using Quadratic residues solve the following congruence relations [2]
 - i. $x^2 \equiv 5 \pmod{11}$
 - ii. $x^2 \equiv 12 \pmod{17}$
(b) Draw the elliptic curve $y^2 = x^3 + ax + b \pmod{r}$, where $a = 3$, $b = 5$, and $r = 13$. [3]
Add two points (4,9) and (12,1).
(c) Using the Rabin cryptosystem with $p=47$ and $q=11$, encrypt the plain text $P=17$ to find the cipher text. Use the Chinese Remainder Theorem to find the four possible plain texts. [5]
4. (a) Write an algorithm in pseudocode to find QR and QNR for any Z_p^* . [2]
(b) Show the encryption and decryption diagram for ECB mode when Cipher-text stealing is used. [3]
(c) Discuss OFB mode with suitable diagram and list its advantages and disadvantages. [5]
5. (a) Define a state in AES. How many states are there in each version of AES? [2]
(b) What is steganography? Explain the LSB based steganography technique. [3]
(c) Explain the Diffie Hellman Key exchange protocol with suitable diagram. How ECC can be used to generate the shared secret key, explain with example. [5]

*****Wish you all the best*****